Customer Churn Prediction - API Deployment Proof

1. Deployment Architecture

Stack Used

- FastAPI (Inference Layer)
- Scikit-learn (Logistic Regression)
- Joblib (Model Persistence)
- Docker (Containerization)
- Uvicorn (ASGI Server)

Container Command Used

```
docker run -p 8000:8000 churn-api
```

2. **API Endpoint Test (Swagger UI Screenshot)**

API Test Details

- Endpoint: POST /predict
- Input: Feature vector (preprocessed customer attributes)
- Decision Threshold: 0.4
- Output:
    - prediction: 0
    - churn_probability: 0.0072
- HTTP Status: 200 OK

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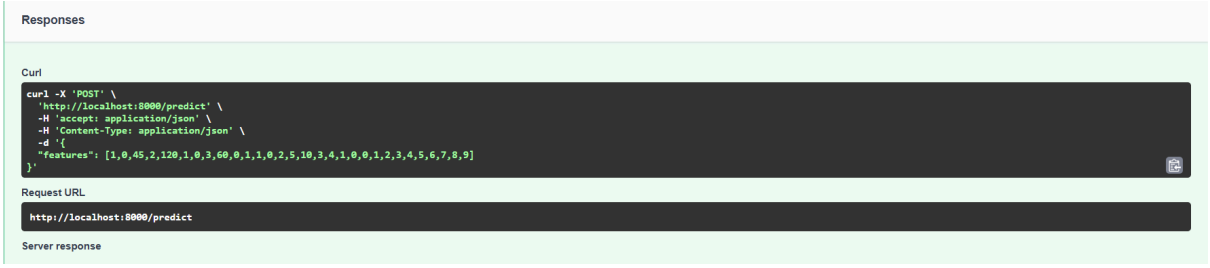
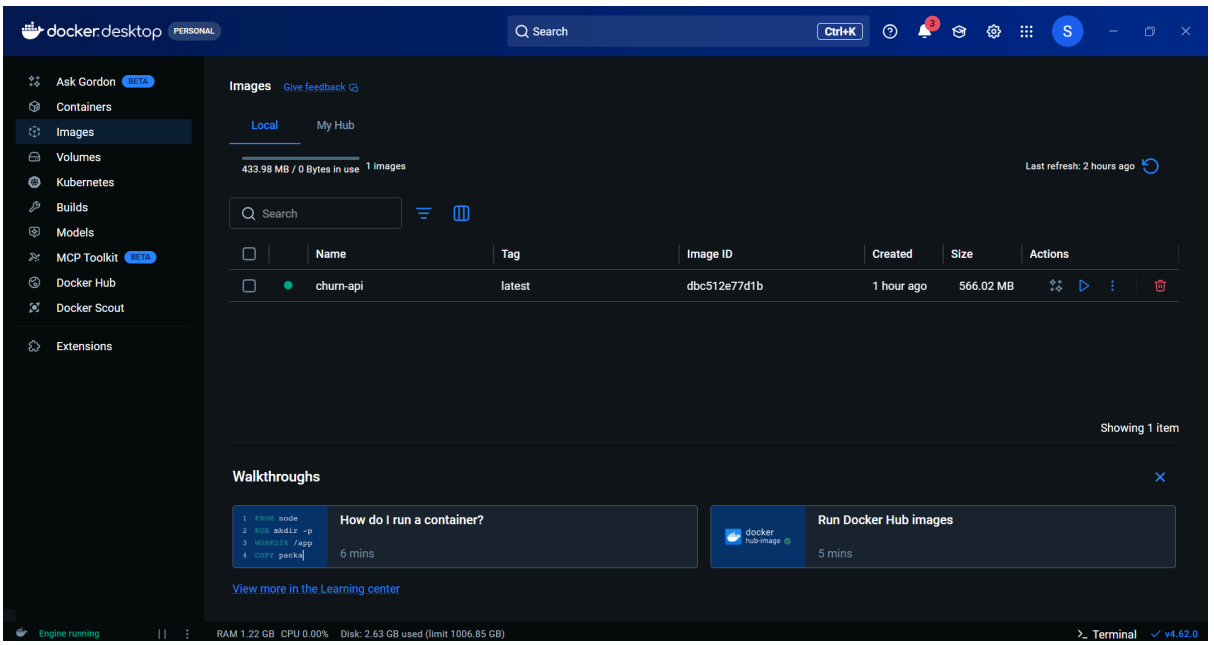
3. cURL Verification



This shows:

- It works outside Swagger
- It's a real API endpoint

4. Deployment Confirmation



- Running container
- Port mapping 8000:8000
- Image name churn-ap